UNFIRE / ENERGY LENS

Energy decisions. Made clear.

Digital Solutions | Residential energy affordability | Solo entry

PROBLEM AND USER

Households and community advisors need to compare hypothetical energy costs without opaque savings promises or unauthorized account actions. The paying customer for a future deployment is not identified yet.

DEMONSTRATED PROTOTYPE

Browser-local estimator: editable synthetic kWh, peak/off-peak rates and shifted demand. Clinic safety preset plus deterministic approval/deny policy. Ten local test assertions. No paid API, personal data, utility connection or payment execution.

MEASURED SYNTHETIC EXAMPLE

420 kWh/month and hypothetical rates: $94.08 baseline, $87.02 modeled case; $7.06 modeled difference (about 7.5%). Neither rates nor savings are verified in a real household.

UNIT ECONOMICS, STATED AS ASSUMPTIONS

Demonstration API cost: $0 (excludes developer labor, hardware and hosting). Illustrative only: a proposed $2,100 three-month pilot across 100 households would equal $21 per household, assuming $1,200 setup and $300/month operations. Not quoted, funded or validated.

6 TO 18-MONTH PLAN

0-3 months: user testing, accessible interface and tariff source permission. 3-6: one dated licensed tariff source and independent model validation. 6-18: consented nonprofit pilot; no critical-load automation.

RISKS AND DISCLOSURE

Tariff eligibility, demand-shift realism, taxes, local regulation, accessibility and actual costs are unverified. Prototype, source and 86s demo pre-date Oct 30 and are prior work; no field deployment or generative-AI integration is claimed.